

Laiba Afreen

Forward Propagation

Given values:

Inputs: $x_1 = 0.5, x_2 = 0.0$

$$W_1 = \begin{bmatrix} 0.1 & 0.2 \\ 0.3 & 0.4 \end{bmatrix}, b_1 = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$W_2 = \begin{bmatrix} 0.5 & 0.6 \end{bmatrix}, b_2 = 0$$

Step 1: Hidden neuron 1's linear value ($z_{1,1}$)

$$\begin{aligned} z_{1,1} &= (0.1 \times 0.5) + (0.2 \times 0.0) + 0 \\ &= 0.05 \end{aligned}$$

Step 2: Hidden neuron 2's linear value ($z_{1,2}$)

$$\begin{aligned} z_{1,2} &= (0.3 \times 0.5) + (0.4 \times 0.0) + 0 \\ &= 0.15 \end{aligned}$$

Step 3: Squash both with Sigmoid ($a_i = \sigma(z_i)$)

$$\text{Formula: } \sigma(z) = \frac{1}{1 + e^{-z}}$$

$$a_{1,1} = \sigma(0.05) = \frac{1}{1 + e^{-0.05}} = \frac{1}{1 + (2.71828)^{-0.05}}$$

$$= 0.5125$$

$$a_{1,2} = \sigma(0.15) = \frac{1}{1 + e^{-0.15}} = \frac{1}{1 + (2.71828)^{-0.15}} = 0.5374$$

Step 4: Output layer linear combination:

$$z_2 = W_2 \cdot a_1 + b_2$$

$$\begin{aligned} z_2 &= (0.5 \times 0.5125) + (0.6 \times 0.5374) + 0 \\ &= 0.2563 + 0.3224 \\ z_2 &= 0.5787 \end{aligned}$$

Step 5: Final Prediction ($\hat{y} = \sigma(z_2)$)

$$\begin{aligned} \hat{y} &= \sigma(0.5787) = \frac{1}{1 + e^{-0.5787}} = \frac{1}{1 + (2.71828)^{-0.5787}} \\ \hat{y} &= 0.6408 \end{aligned}$$

Step 6: Loss (How wrong the prediction was)

Formula: $L = -y(\log(\hat{y}) - (1-y)\log(1-\hat{y}))$

Since $y=1$, then second part become 0:

$$L = -\log(0.6408) = 0.4451$$